

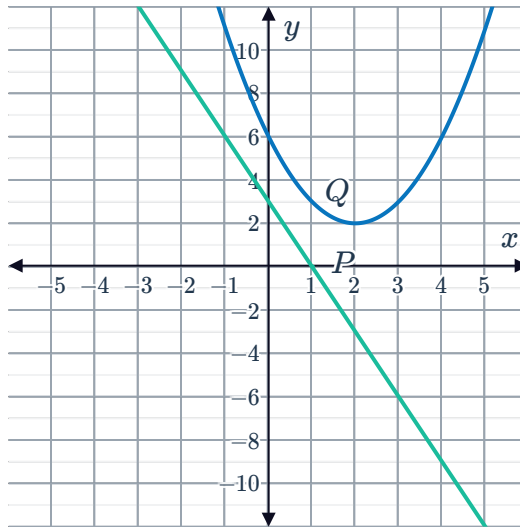
Compare function types



1

a Function P is a line and function Q is a parabola.

b

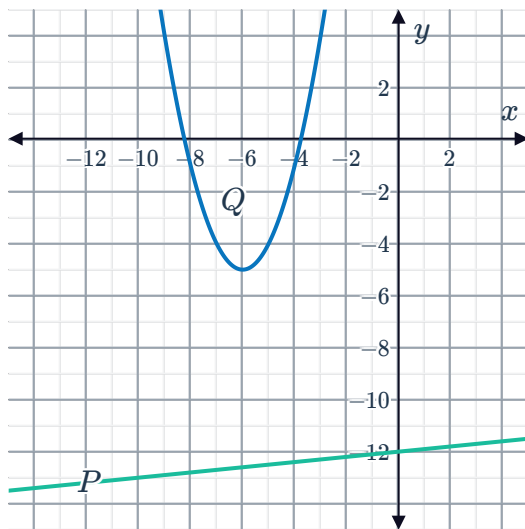


c Function Q

d Functions P and Q are decreasing.

2

a



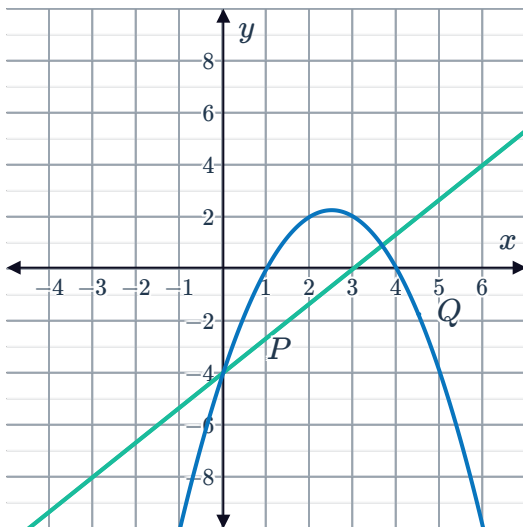
b Zero times

c Function Q

d Function Q

3

a

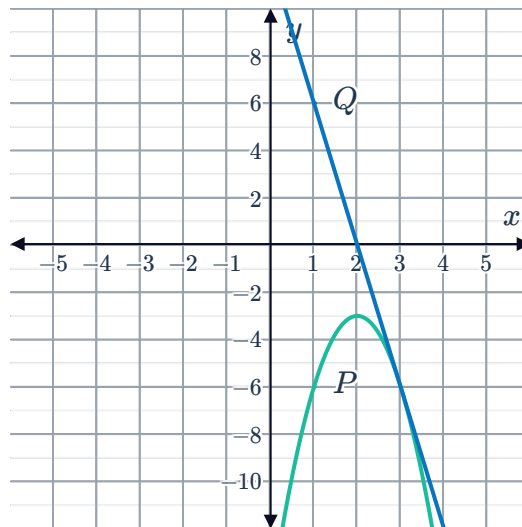


b Twice

c They are equal at $x = 0$.

4

a



b Once

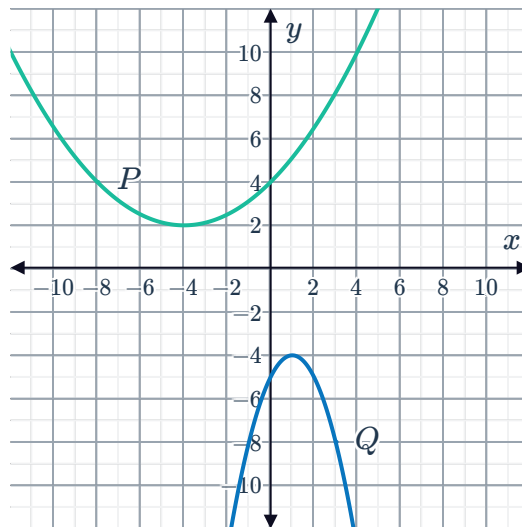
c Function Q

d Function P is increasing and function Q is decreasing.

5 Function P

6

a



b Function P

c Function P

d Parabola P and parabola Q are increasing.

7

a Function A

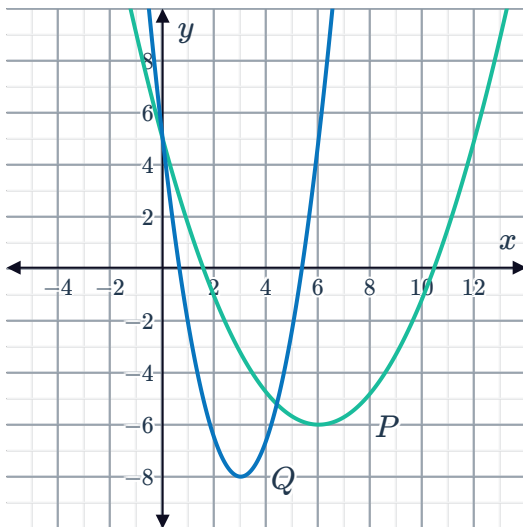
b Function B

c 2

d 0

8

a



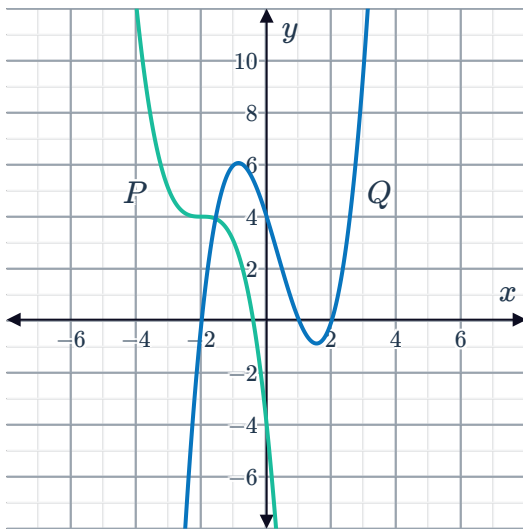
b Parabola Q

c Twice

d Parabola Q

9

a



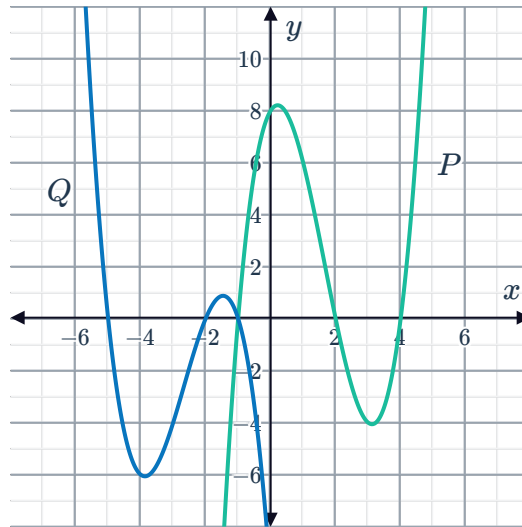
Cubic function Q

c 1

d 3

10

a

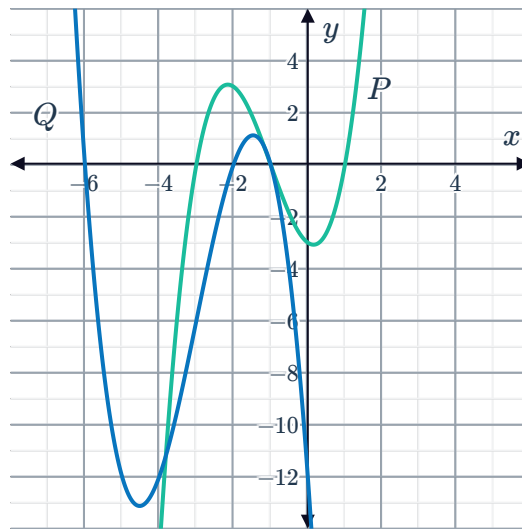


b Function P is decreasing and function Q is decreasing.

c As x approaches ∞ , function P approaches ∞ and function Q approaches $-\infty$.

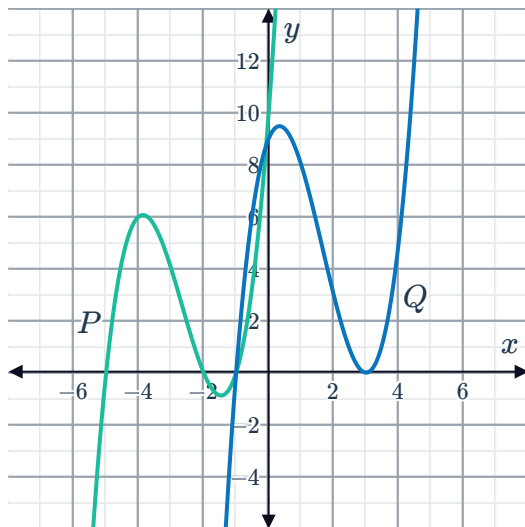
11

a

b Function P c Function P is increasing and function Q is increasing.d As x approaches ∞ , function Q approaches $-\infty$ and function P approaches ∞ .

12

a

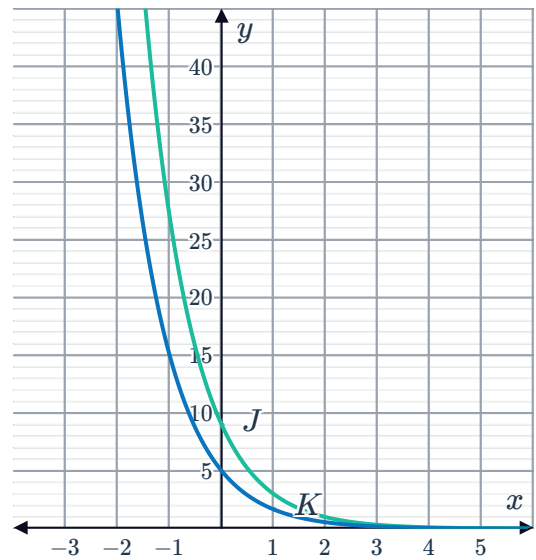
b Cubic function P

c 3

d 2

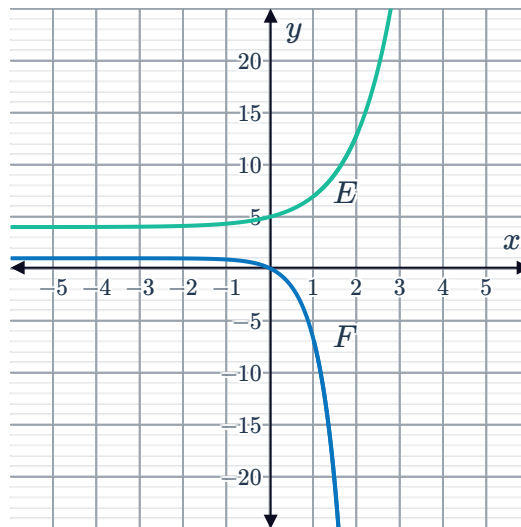
13

a Decreasing

c As $x \rightarrow \infty$, functions $J, K \rightarrow 0$.d Dilate $y = 3^{-x}$ vertically by a factor of 5.

14

a



b Zero times

c As $x \rightarrow \infty$, the function $E \rightarrow \infty$ and function $F \rightarrow -\infty$.

15

a Once

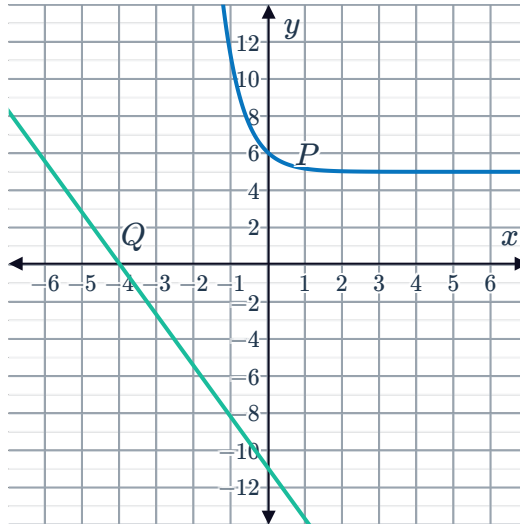
b Function P

16



a P is an exponential function and Q is a linear function.

b

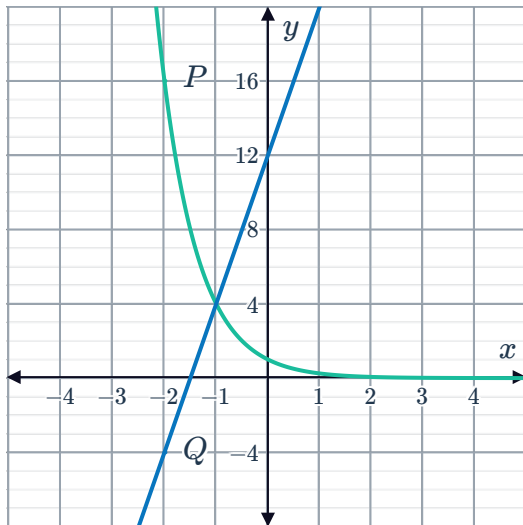


c Function P

d Zero times

17

a



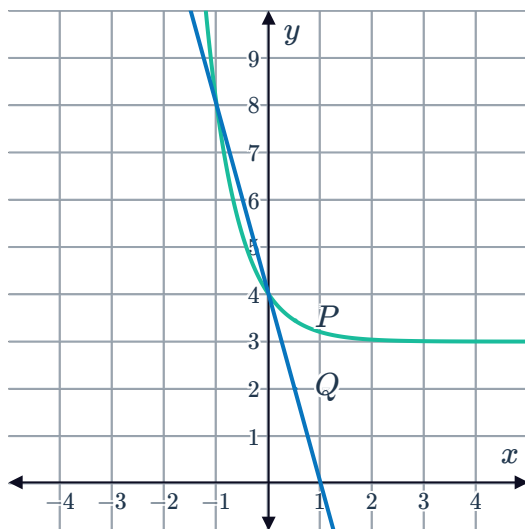
b Once

c Function Q

d Function Q

18

a



b Twice

c Function P

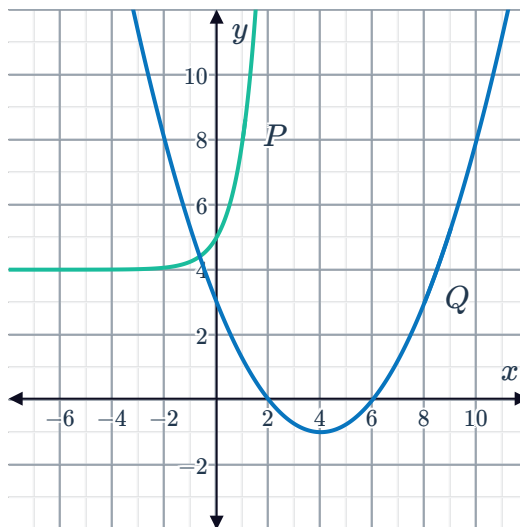
19

a Function A

b Function A is decreasing and function B is increasing.

20

a



b Function P

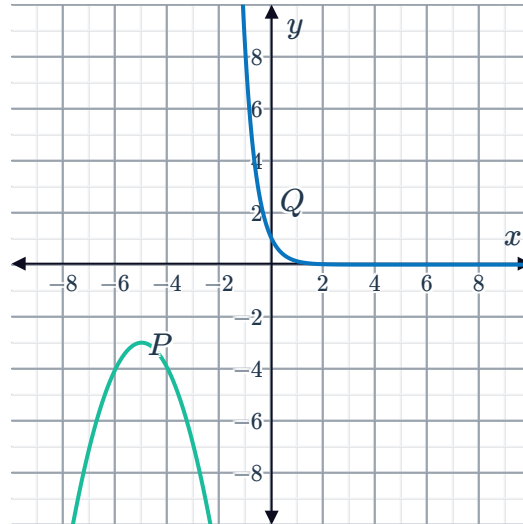
c Function P is increasing and function Q is decreasing.

21



a Function P is quadratic (parabola) and function Q is exponential.

b



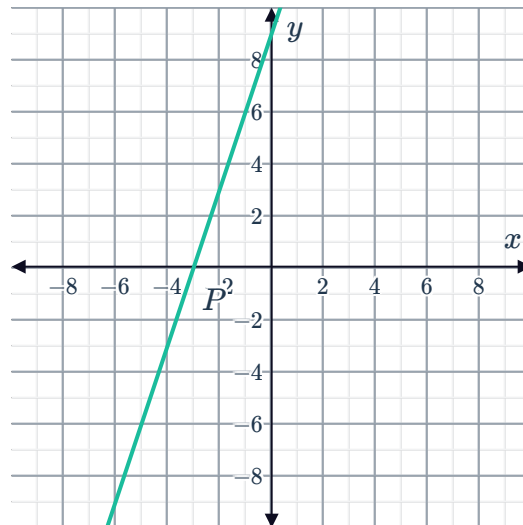
c Zero times

Applications

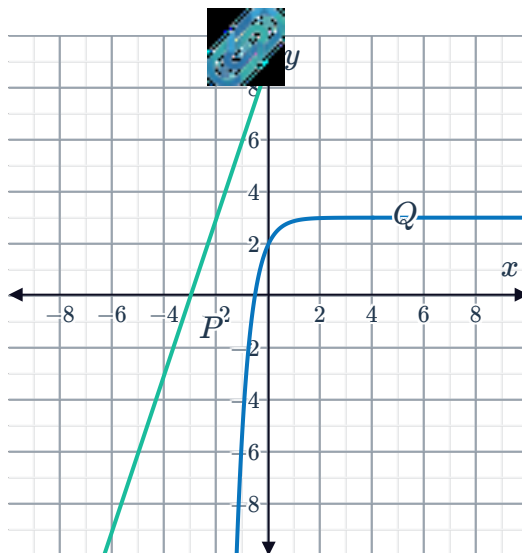
22

a Function P is linear and function Q is non-linear.

b



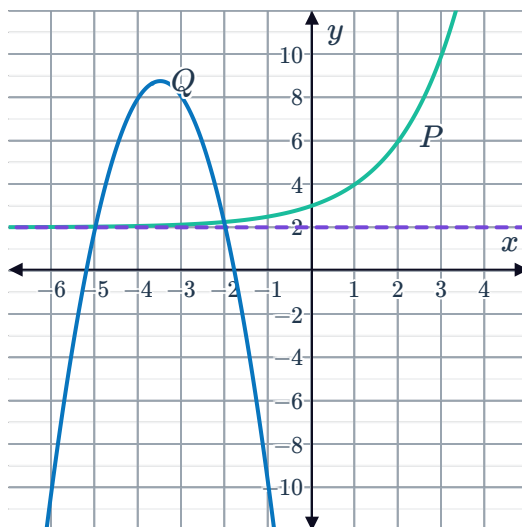
c



d 0

23

a



b As x approaches ∞ , the function Q approaches $-\infty$ while function P approaches ∞ .

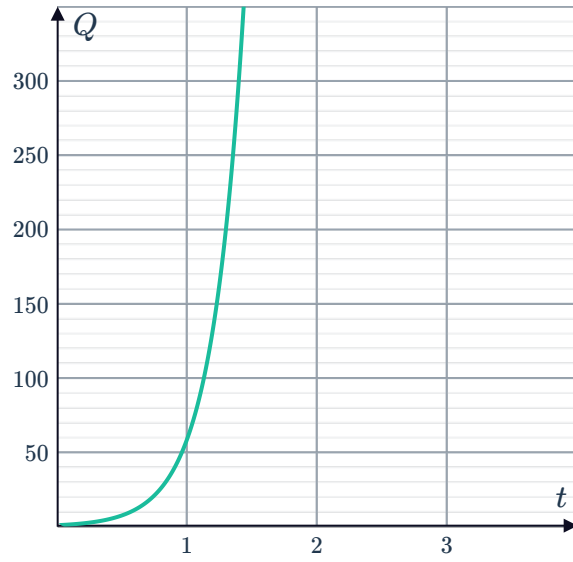
c 2

24



a The population of bacteria J increases faster than the population of K .

b



c

